IS curve. An implication of the above is that any point outside the IS curve shows disequilibrium in the real sector. In Fig. 3.2 we consider two such points 'c' and 'd' which are not on the IS curve, and find out their implications.

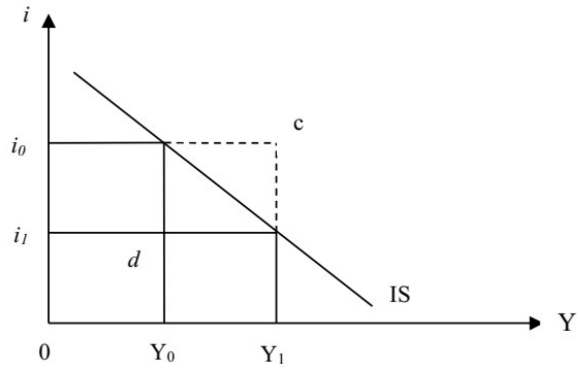

Fig. 3.2: Disequilibrium in the Real Sector

At point 'c' in Fig. 3.2, interest rate is i_0 while output is Y_1 . Notice that at Y_1 level of output, the equilibrium rate of interest should be i_1. Thus, the actual rate interest is higher than what it should. As a result, the aggregate demand will decrease (largely due to decrease in investment) and the rate of interest will decline. It implies that at point-c, there is excess supply of goods in the market. We can generalize that any point which is above and to the right of the IS curve, indicates excess supply of goods.

Let us look at point 'd' which is below and to the left of the IS curve. At point 'd' in Fig. 3.2, income is at Y_0 but the rate of interest is at i_1. Since the rate of interest is lower than the equilibrium rate of interest, the demand for investment will increase. Consequently, the rate of interest will rise upward. Thus, at point 'd' there is excess demand for goods. We can generalize that at any point below and to the left of the IS curve, there is excess demand for goods.

3.2.2 Shift in the IS Curve

The IS curve depicts the relationship between income and interest rate. Therefore, any change in other parameters or variables will result in a shift in the IS curve. The position and the slope of the IS curve depends upon the value of the parameters in equation (3.4). If the IS curve is flatter, the economy is more sensitive to interest rate – a small decrease in interest rate will lead to a large increase in income. On the other hand, a steeper IS curve implies insensitivity to interest rate change.

There are two exogenous variables in equation (3.4), viz., G and T. In equation (3.4) we assumed that government expenditure and taxes are fixed. Thus, changes in taxes and government expenditure will also result in shifts in the IS curve. Let us look into equation (3.4). An increase in the value of $\bar{G}$ will result in an increase in the intercept; thus, there will be an upward parallel shift in the IS curve. An increase in $\bar{T}$, on the other hand, will decrease the intercept and shift the IS curve downward to the left.

A change in the value of a parameter leads to a shift in the IS curve. An increase in autonomous consumption (the coefficient 'a' in equation (3.4)) or autonomous investment (the coefficient 'c' in equation (3.4)) will lead to a parallel shift of the IS curve upward to the right. A change in the marginal propensity to consume, however, will affect the slope of the IS curve.

Check Your Progress 1

1) What will be the nature of change in the IS curve if there is a decrease in the propensity to consume?

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2) What is the implication of the following on the position of the IS curve?

(i) Decrease in autonomous government expenditure

(ii) Decrease in tax rate

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3.3 EQUILIBRIUM IN THE MONETARY SECTOR

The derivation of LM curve utilises the Keynesian view that speculative demand and transaction demand for money are separate. The speculative demand for money depends on the rate of interest while the transaction demand and the precautionary demand for money depend on the level of income. Thus, total demand for money (M) depends on both income (Y) and rate of interest (i). The demand for money in real terms ($\frac{M^d}{P}$) is given by

$$L = kY - hi \qquad \ldots (3.5)$$

where $k > 0$, $h > 0$; L is the demand for real balances (liquidity), Y is aggregate income and i is the rate of interest. You should remember three issues about the monetary sector: (i) A person would like to deposit his money in bank and earn interest on it. The alternative is to hold on to the cash, where no interest income is earned. Thus, the opportunity cost of holding money is the interest foregone. If there is an increase in the rate of interest, the interest foregone increases. Conversely, if the rate of interest decreases, the interest income foregone decreases. Therefore, the demand for money is a downward-sloping curve, if we measure rate of interest on the y-axis and quantity of money on the x-axis. You may be familiar with the concept of liquidity trap; a situation where the rate of interest is very low, and people prefer to keep all their money in the form of liquid cash as the interest income foregone is negligible. (ii) The demand for real balances is described for a given level of income. We consider the money demand function while keeping the level of income fixed (say, $L(Y_0)$). If there is an increase in the level of income, there is an increase in the level of transactions and wealth in the economy, which in turn increases the demand for money. Thus, there will be a parallel shift in the money demand function. (iii) The supply of money is determined by the central bank and it is constant. Thus, the supply of money is a vertical straight line. If the supply of money is increased by the central bank, the vertical straight line will shift to the right. When we deal with the short-run, we assume that money supply does not change.

3.3.1 Derivation of the LM Curve

The supply of money in real terms is given by $\frac{M^s}{P}$. Equilibrium in the money market is achieved when $L = M = \bar{M}$. Thus, equilibrium in the money market is given by

$$\frac{\bar{M}}{\bar{P}} = kY - hi \qquad \ldots (3.6)$$

We can re-arrange terms in the above equation and obtain

$$Y = -\frac{1}{h}\frac{\bar{M}}{\bar{P}} + \frac{h}{k}i \qquad \ldots (3.6a)$$

There is a positive relationship between Y and i in the LM curve. In order to find out the slope of equation (3.6) you have to express the equation in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the LM curve as $\frac{k}{h}$. Depending upon the value of the parameters, the LM curve would be flatter or steeper.

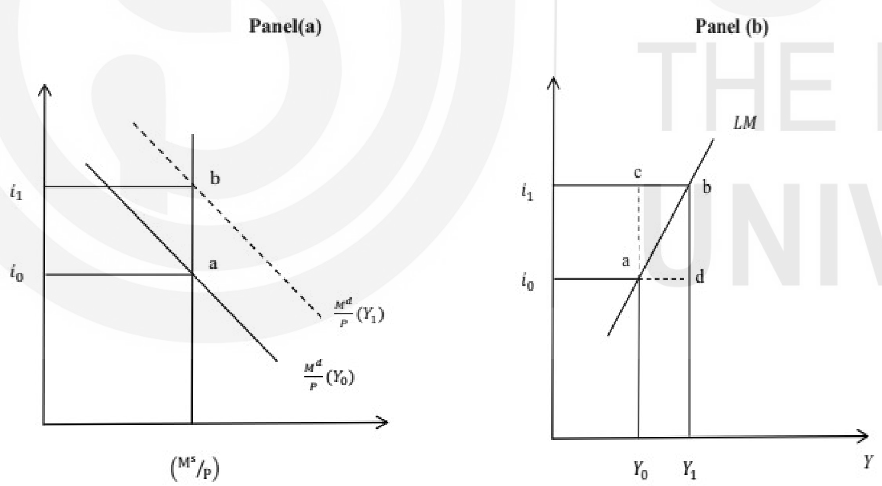

Fig. 3.3: Derivation of the LM Curve

In Fig. 3.3, Panel (a) we depict the equilibrium in the money market (or, in the monetary sector). We take demand for real balances $(\frac{M}{P})$ on the x-axis and the rate of interest on the y-axis. Let the money demand in the economy be given as $\frac{M^d}{P}(Y_0)$ when the aggregate income is Y_0. In panel (a) of Fig. 3.3, this equilibrium is represented by point ‘a’. The equilibrium rate of interest and income are i_0 and Y_0 w respectively.

Let there be an increase in the level of income from Y_0 to Y_1. Such an increase in income will lead to a parallel upward shift in the money demand function. We depict this money demand function as a dotted line $\frac{M^d}{P}(Y_1)$ in Panel (a) of Fig. 3.3. With money supply remaining constant, an increase in aggregate demand will lead to a rise in the interest rate. The economy is now at equilibrium with output level Y_1 and interest rate i_1. The intersection between the money supply function and the money demand function is denoted by point 'b' in Fig. 3.3, panel (a).

In panel (b) of Fig. 3.3 we plot the LM curve by taking combinations of interest rate and aggregate income where the money market is in equilibrium. We have two such points, 'a' and 'b' in panel (a) of Fig. 3.3. You should note that the LM curve is the combination of all the points where the money market is in equilibrium. The LM curve is upward-sloping as you can see from the figure. An implication of the LM curve is that as income increases people want to hold more money, thus driving up the interest rate. If the speculative demand for money is perfectly elastic to interest rate (liquidity trap situation) the LM curve will be horizontal. On the other hand, if speculative demand is perfectly inelastic to interest rate, the LM curve will be vertical.

To the left side of the LM curve there is excess supply of money (for example, point 'c' in Fig. 3.3 panel (b)). Excess supply in the money market implies that the demand for money is less than its supply. It means that people want to reduce the amount of liquid cash they possess. This induces people to buy bonds so that bond prices rise and rate of interest declines. A decline in the rate of interest restores equilibrium in the economy. To the right side of the LM curve, there is excess demand for money (for example, point 'd' in Fig. 3.3 panel (b)). Excess demand for money causes people to sell bonds (so that they can keep higher amount of liquid cash). This leads to a fall in bond prices and rise in the rate of interest. This way equilibrium in the monetary sector is restored.

3.3.2 Shift in the LM Curve

The position and the slope of the LM curve depend upon certain factors, including money supply. An increase in money supply shifts the LM curve to the right. As a result, at every level of output, the equilibrium rate of interest is lower than before. Thus, increase in money supply shifts the LM curve to the right. Conversely, a decrease in money supply will shift the LM curve to the left so that equilibrium rate of interest is higher at every level of output.

3.4 SIMULTANEOUS EQUILIBRIUM OF REAL AND MONETARY SECTORS

Now let us combine IS and LM curves as shown in Fig. 3.4. Such integration of IS-LM gives a unique combination of i and Y, which represents equilibrium in both real market and money market. In Fig. 3.4 we superimpose IS_0 and LM_0 in the same diagram. We measure income (Y) on the x-axis income (Y) and rate of interest on the y-axis. The equilibrium rate of interest and the equilibrium output level are given by i_0 and Y_0 respectively.

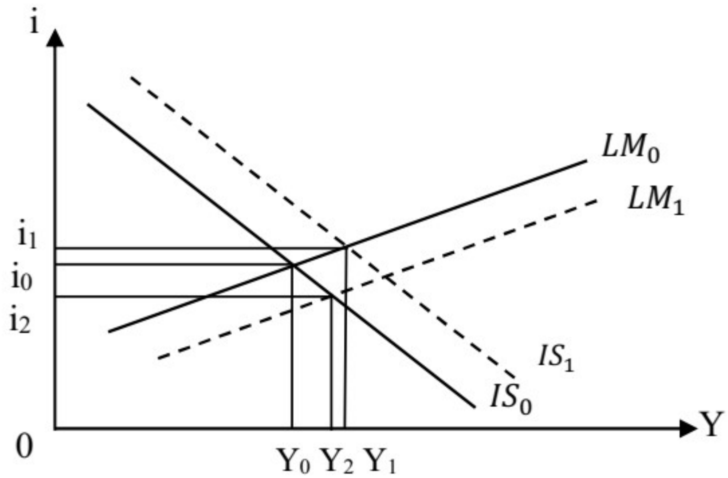

Fig. 3.4: Simultaneous Equilibrium

3.4.1 Shocks in the IS-LM Model

Previously we have discussed about the factors that lead to a shift in the IS curve and the LM curve. Often the economy experiences exogenous demand shocks and supply shocks. Let us analyse the impact of such shocks in terms of the IS-LM model. Suppose there is a boom in the stock market, which changes the wealth of households. Such a situation will increase the level of consumption in the economy. Consequently, there will be an increase in aggregate expenditure and the IS curve will shift to the right from IS_0 to IS_1 (see Fig. 3.4). As a result, there is an increase in output from Y_0 to Y_1. A negative shock such as a decrease in business confidence of firms will shift the IS curve to the left. Such a left-ward shift in the IS curve will decrease output as well as interest rate.

Let us consider another example. Suppose there is a series of online financial fraud cases in the country. This will send a negative signal to people – this will discourage people to keep their income in the formal system; there will be a decline in online transactions and the demand for money will increase. As a result, the LM curve will shift to the right from LM_0 to LM_1(see Fig. 3.4). Consequently, there will be an increase in output from Y_0 to Y_2, and decrease in interest rate from i_0 to i_2.

You should note that the nature and extent of change in output and interest rate will depend upon the slope of the IS and LM curves. If the LM curve is flatter, monetary policy is quite effective. A small decrease in interest rate by the central bank will increase output by a large amount. If the LM curve is steeper, monetary policy will not be effective. In order to increase output by a small amount we will need a large decrease in the interest rate.

3.4.2 Adjustment Process in the Economy

A question arises at this point: What happens to the economy if it is operating at a point which is not at the intersection of the IS and LM curves? Does it automatically converge to the equilibrium point? Let us try to answer this question.

If the economy is not operating at the unique point where the IS and LM curves intersect, there is disequilibrium in the economy. In that case an adjustment process starts and the economy moves to an equilibrium point. The following two issues are important in this context:

a) If there is an excess demand for money (to the right of the LM curve), the interest rate will increase. The interest rate will fall in case we observe excess supply in the money market. Fall in interest rate will lead to increase in investment. Increase in investment will affect the IS curve.

b) Excess demand for goods (to the left of the IS curve) leads to an increase in output. Increase in output will lead to increased demand for money. Increase in demand for money will lead to rise in interest rate and outward shift in the LM curve.

We observe from the above that both real sector and monetary sector are inter-related. A real shock to the economy will have effects on the monetary sector and *vice versa*.

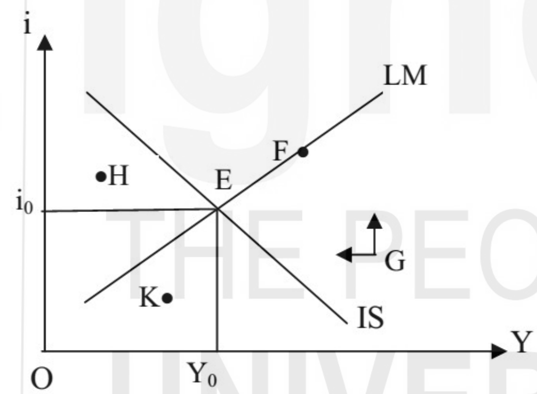

Fig. 3.5: Convergence to Equilibrium Point

In Fig. 3.5 we plot the IS and LM curves. The economy is in equilibrium at point E, with i_0 to Y_0 as equilibrium interest rate and output respectively. Suppose, due to certain shock to the economy, the equilibrium is disturbed. We have indicated the points F, G, H and K where the economy is not in equilibrium.

Let us consider point G. As mentioned earlier, at point G, there is excess demand for money and excess supply of goods. The arrows given at point G specify the direction in which adjustment takes place. Due to excess demand for money, there will be a rise in the rate of interest. This is because people will sell bonds so as to hold more of liquid cash, as a result of which the bond prices will decline and rate of interest will increase. Upward pointing arrow in Fig. 3.5 represents rising interest rate. At point G, there is excess supply of goods also. This will lead to involuntary accumulation of goods; inventories will increase. As a result, firms will cut down production and the level of output will decline. The leftward pointing arrow represents the declining output. Finally, the economy will move towards equilibrium point E.

Points F, H and K in Fig. 3.5 also are disequilibrium points. You can analyse the supply and demand conditions at these points and find out the directions in which adjustment will take place. For example, at point 'H' there is excess supply of money (since it is to the left of the LM curve) and excess demand for goods (since it is to the left of the IS curve). Excess supply of money will lead to a decrease in interest rate. Excess demand for goods will lead to an increase in output level. Consequently, the economy will move towards the equilibrium point E.

Suppose the economy is at point F (see Fig. 3.5) – here the monetary sector is in equilibrium, but the real sector is in disequilibrium. There is an excess supply of goods in the economy. This will lead to an increase in inventory accumulation. As a result, firms will decrease their production levels. Decrease in output will lead to a decrease in the demand for money and consequent decrease in the rate of interest. Finally, the economy will reach the equilibrium point E where both the IS and LM curves intersect.

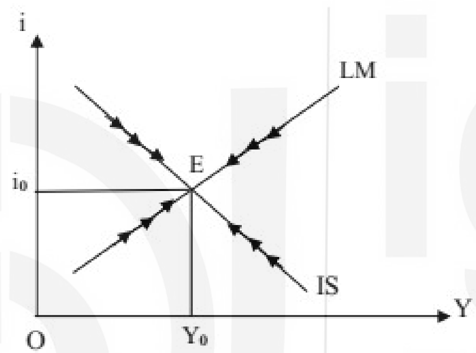

Fig. 3.6: Adjustment Process in the Economy

Note that when one of the two markets is observing disequilibrium, the economy reaches back to equilibrium after the adjustment process. In Fig. 3.6 we have indicated the direction of change by arrow marks.

3.4.3 Impact of Supply Shocks

In the IS-LM model it is not necessary that the equilibrium occurs at the level of potential output of the economy (or, full employment output). Equilibrium could be at any level of output in the economy. An objective of policy makers has always been that the economy should operate at potential output level with unemployment at its natural level. In the long run does the economy maintain equilibrium at the full employment output level?

In the long run prices do not remain constant. If actual output is above the potential out, there will be price rise. The aggregate expenditure of the economy (such as consumption, investment and government expenditure) describes the demand for goods and services, which refer to physical quantities of goods and services. A change in the price level may not affect these quantities. Thus, the IS curve is relatively stable – it may not shift. The LM curve, on the other hand, is based on the supply of and demand for real balances. Thus, a change in price level will shift the LM curve.

The full employment output is given by the production capacity of the economy. It is derived from the production function (see Unit 1). Suppose the full employment output (potential output) of the economy is given by Y^*. We depict it by a vertical straight line in Fig. 3.7. The IS_0 curve and LM_0 curve intersect at E which corresponds to the full employment output.

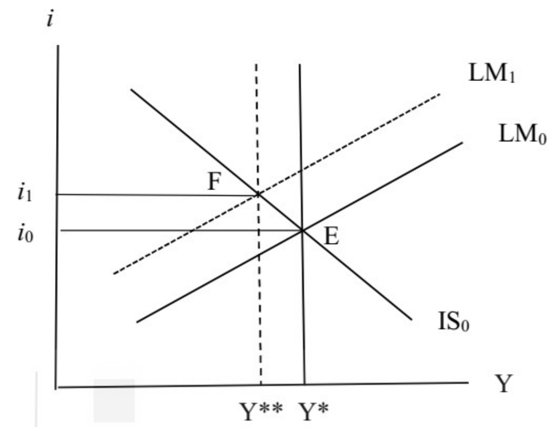

Fig. 3.7: Adjustment to Adverse Supply Shocks

Suppose the economy gets an adverse supply shock (for example, decline in agricultural productivity due to drought), as a result of which there is a decrease in the potential output of the economy from Y^* to Y^{**}. As pointed out above, the IS curve is relatively stable in such situations and the economy adjusts through a shift in the LM curve. Due to the adverse supply shock, there is a rise in prices and consequent leftward shift in the LM curve. Thus, there is a rise in the rate of interest and the new equilibrium is at F.

Check Your Progress 2

1) Explain how the LM curve is derived.

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2) Suppose the LM curve is vertical. What are its implications?

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3) In the IS-LM model, explain how adjustment to supply shocks takes place.

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3.5 AD-AS MODEL

In the IS-LM model we assumed that price level is constant. Let us relax that assumption and look into the impact of prices on equilibrium output in the economy. First, we derive the aggregate demand (AD) curve. Subsequently, we juxtapose the AD curve with the aggregate supply (AS) curve and find out the equilibrium level of output.

3.5.1 Derivation of the AD Curve

A change in the price level changes the real money supply ($\frac{M^S}{P}$) in the economy. This will result in a shift in the LM curve if money supply (M) is assumed to be constant. Thus, there will be a change in the equilibrium rate of interest and output. If we assume that the price level keeps changing over time, we will obtain a series of equilibrium levels of output corresponding to different price levels. If we plot such combinations of prices and output, we obtain the aggregate demand (AD) curve.

The derivation of the AD curve from IS-LM curves is shown in Fig. 3.8. In panel (a) of Fig. 3.8 we juxtapose IS_0 and LM_0 curves intersecting at point E with an equilibrium income level of Y_0 and interest rate i_0. The LM_0 curve is based on real money stock of M_0/P_0. Thus, we describe the combination P_0 and Y_0 as point 'a' on the *AD* curve in panel (b) of Fig.3.8.

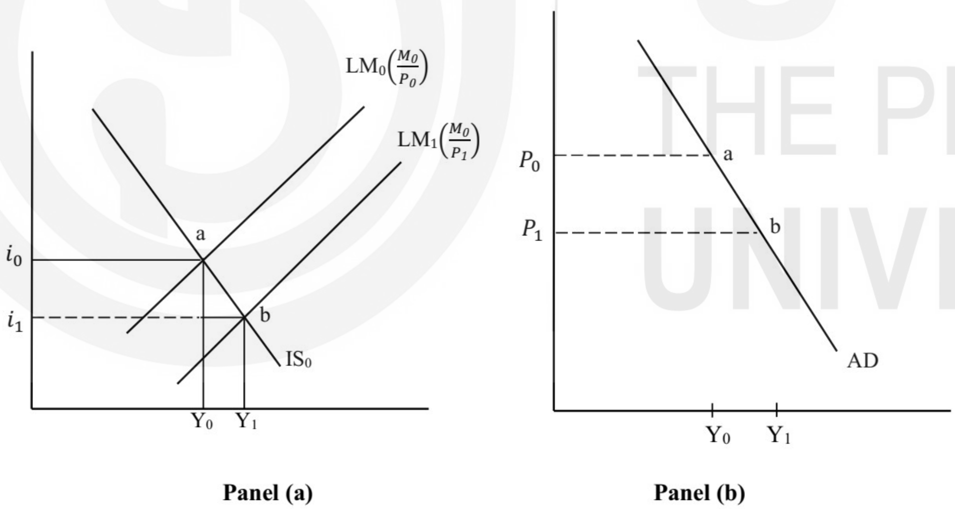

Fig. 3.8: Derivation of AD Curve

Now, let us assume that price level declines to P_1 . If there is no change in money stock, supply of real balances increases to M_0/P_1. Note that the increase in real balances is due to decrease in prices, not due to an increase nominal money stock. The *LM* curve shifts to the right. In panel (a) of Fig. 3.8 we depict the new LM curve as LM_1. With IS_0 and LM_1, the new equilibrium point is 'b'; the rate of interest falls to i_1 and income rises to Y_1. The combination of P_1 and Y_1 becomes point 'b' in Panel (b) of Fig. 3.7. By joining all such points as 'a' and 'b', we obtain the AD curve.

The AD curve is downward-sloping, which indicates an inverse relationship between prices and output. A decline in price level increases the real money supply, which creates disequilibrium in the money market. Increase in the supply of real balances lead to a decrease in interest rate. A decrease in the rate of interest restores equilibrium in the money market. A decrease in the rate of interest leads to an increase in the level of investment. An increase in the level of investment leads to an increase in aggregate expenditure, thereby increasing the level of output.

The AD curve is flatter, if a given change in price leads to a larger change in output. On the other hand, the AD curve is steeper if a given change in price leads to a smaller increase in output. The AD curve shifts upward to the right as a result of expansionary fiscal policy (e.g., increase in government expenditure, reduction in tax rate). A contractionary fiscal policy, on the other hand, shifts the AD curve downward to the left. Similarly, an expansionary monetary policy (e.g., reduction in interest rate, increase in money supply through open market operations, reserve requirements, etc.) will shift the AD curve to the right while a contractionary monetary policy will shift the AD curve to the left.

You should note that any factor that leads to a shift in the IS curve or the LM curve to the right will shift the AD curve to the right side. Similarly, any factor that shifts the IS curve or LM curve to the left will shift the AD curve to the left. Thus, an increase in any component of autonomous spending such as an increase in government spending leads to a rightward shift of the AD curve. Also, an increase in the nominal money supply will shift the AD curve upwards.

3.5.2 Aggregate Supply (AS) Curve

From microeconomics we know that the AS curve is upward-sloping – when price of a product increases, firms produce more of the product. In the case of the whole economy, production capacity is limited. Resources available to the economy are not unlimited. The behaviour of the AS curve is different in the long run compared to the short run. Short run, as you know, is a time period in which wages and certain other economic variables do not respond to changes in economic environment. In the long run is a time period in which wages and prices are flexible.

Classical theory discussed in Unit 1 assumed that the AS curve is vertical at the full employment level. Thus, output is perfectly inelastic to changes in prices. Keynes went to the other extreme and assumed that the AS curve is horizontal. An implication of the horizontal AS curve is that firms can increase production to any level without affecting prices. Both the versions (vertical and horizontal) of the AS curve seems unrealistic. As you will learn in later units, the new-classical economists suggest that the AS curve is upward-sloping in the short run. Lucas' in his 'aggregate supply hypothesis' advocate that economic agents respond to price and wage incentives, which influences their trade-off between leisure and work. As a result, when actual wage is higher than equilibrium wage rate, households supply more labour. Conversely, when actual wage rate is lower than equilibrium wage rate, household will decrease their supply of labour. As a result of this, the